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Department of Computer Science and Engineering

Radiosight -Radiology and Insight

**PROJECT SYNOPSIS
OF MAJOR PROJECT**

**BACHELOR OF TECHNOLOGY
Computer Science and Engineering**

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Abstract

Chest radiography remains one of the most accessible and widely used medical imaging techniques for diagnosing pulmonary diseases such as pneumonia, tuberculosis, and COVID-19. However, interpreting chest X-rays requires specialized radiological expertise, which is often scarce in rural or resource-limited healthcare settings. To address this challenge, the proposed Chest Radiology Similarity System leverages advanced deep learning techniques, specifically the EfficientNet architecture, to enable automated analysis and retrieval of visually similar chest X-ray cases from a curated database.

This system goes beyond traditional binary classification by providing comparative insights through similarity retrieval, severity scoring, and disease progression tracking. A key feature is the integration of Grad-CAM (Gradient-weighted Class Activation Mapping), which visually highlights affected lung regions on X-rays, thus enhancing model interpretability and building clinician trust.

The workflow begins with preprocessing steps including image resizing, noise removal, and normalization to improve model performance. The preprocessed images are then fed into the EfficientNet model for feature extraction and similarity computation. Retrieved cases and severity metrics are presented through a scalable and user-friendly web interface developed with Next.js, which enables seamless access and interaction across devices.

The system's feasibility is supported by the availability of large annotated datasets such as CheXpert and ChestX-ray8, and the computational efficiency of EfficientNet allows deployment even on mid-range hardware. By enabling transparent, accurate, and accessible chest X-ray analysis, this project aims to bridge the diagnostic gap in underserved regions, reduce reliance on scarce radiologists, and ultimately contribute to improved pulmonary healthcare outcomes.

Introduction

Chest radiography is one of the most commonly performed diagnostic imaging procedures worldwide due to its non-invasive nature, low cost, and accessibility. Chest X-rays provide crucial visual information about the heart, lungs, airways, blood vessels, and bones within the chest cavity. They are frontline tools in diagnosing and monitoring a wide range of pulmonary and cardiovascular conditions, such as pneumonia, tuberculosis, chronic obstructive pulmonary disease, lung cancer, and COVID-19. The images produced by chest X-rays allow clinicians to assess anatomical and pathological changes, aiding in timely treatment decisions.

Despite the widespread use of chest radiography, accurate interpretation of X-ray images demands specialized radiological expertise. This expertise is often scarce or inconsistent in rural, semi-urban, or resource-limited healthcare settings, leading to delays in diagnosis and treatment. The reliance on human expertise also introduces variability stemming from differences in experience and potential fatigue. Furthermore, subtle abnormalities or progressive disease changes may be overlooked or misinterpreted using conventional radiological assessments.

Recent advances in artificial intelligence (AI) and deep learning have demonstrated great potential in automating and enhancing the analysis of medical images. AI models can identify intricate patterns and features in chest X-rays, facilitating objective, accurate, and reproducible diagnoses. Beyond simple binary classifications, AI systems are increasingly capable of retrieving visually similar cases from large databases, enabling comparative analysis and incremental decision support. This approach empowers clinicians with contextual insights that can improve diagnostic confidence and patient outcomes.

The proposed Chest Radiology Similarity System leverages EfficientNet, a cutting-edge convolutional neural network architecture, to automatically analyze and extract meaningful features from chest X-ray images. By computing similarities with a curated database, the system retrieves relevant comparable cases. Additionally, it incorporates Grad-CAM (Gradient-weighted Class Activation Mapping) for visual explainability, highlighting affected lung regions to facilitate clinician interpretation. The integration of severity scoring and disease progression tracking enables longitudinal patient evaluation, supporting personalized and adaptive clinical management.

Accessible through a modern, scalable Next.js web interface, the system is designed for easy deployment in diverse healthcare environments, including clinics with limited radiology resources. This technology aims to bridge the gap between advanced diagnostic tools and under-resourced settings, ultimately enhancing healthcare equity and improving pulmonary disease diagnosis and monitoring globally.

Rationale

Chest radiography is a fundamental diagnostic tool broadly used in medical practice to detect and monitor lung and heart conditions due to its accessibility, rapid acquisition, and cost-effectiveness. However, the accurate interpretation of chest X-rays is a complex and challenging task even for trained radiologists. The difficulties arise from various factors including overlapping anatomical structures, subtle imaging findings, and similarity of radiographic presentation across different pulmonary diseases. These challenges lead to variability in diagnostic accuracy and can delay appropriate patient management.

A critical issue in many regions, especially in developing countries and rural or semi-urban healthcare settings, is the severe shortage of qualified radiologists. For example, in some areas the ratio of radiologists to population can be as low as one per several hundred thousand people. Consequently, many X-ray images remain unanalyzed or are interpreted by less experienced personnel, leading to delayed diagnoses, misinterpretations, and suboptimal treatment decisions. This shortage is further exacerbated by increasing demand due to rising population and a growing burden of pulmonary diseases.

Traditional chest X-ray interpretation methods also suffer from subjectivity and inter-reader variability. Studies have shown discrepancies among radiologists in detecting certain lung abnormalities, which can impact patient outcomes. Moreover, the interpretation workload can cause fatigue and errors, further undermining diagnostic reliability.

In this context, artificial intelligence (AI) and deep learning technologies offer promising solutions to augment radiological services. Automated AI systems can analyze large volumes of chest X-rays rapidly and consistently, identifying patterns that might be missed by human observers. Beyond simple classification, AI models that retrieve visually similar cases can provide valuable contextual information to clinicians, improving confidence and decision-making.

However, existing commercial AI tools often have significant limitations: they can be prohibitively expensive, lack transparency, and typically provide only binary presence/absence results without detailed explainability or progression insights. This limits their utility in underserved and resource-constrained environments.

The proposed Chest Radiology Similarity System addresses these gaps by combining highly efficient deep learning (using EfficientNet) with explainable AI techniques (such as Grad-CAM) and similarity-based case retrieval. This approach enhances interpretability, facilitates clinical insight through severity scoring and longitudinal disease tracking, and democratizes access via a scalable web platform. It empowers healthcare providers in low-resource settings to leverage advanced diagnostic support, reduce dependence on scarce radiology expertise, and enable timely and accurate pulmonary disease management.

Objectives

The primary goal of this project is to harness the power of artificial intelligence to enhance the accuracy, interpretability, and accessibility of chest X-ray diagnostics. By leveraging advanced deep learning models and explainable AI techniques, the system aims to support clinicians in making well-informed decisions. The following objectives outline the key targets for developing a robust, efficient, and user-centric diagnostic solution

- Develop an **AI-powered** similarity-based chest X-ray diagnostic system using **EfficientNet**, capable of accurately analyzing and retrieving visually similar cases from large medical imaging databases.
- Integrate **Grad-CAM** for explainability to highlight affected lung regions on X-rays, providing transparent visual explanations that aid clinicians in interpreting AI-driven results.
- **Implement severity scoring and disease progression analysis** to enable quantitative assessment and longitudinal tracking of pulmonary conditions, supporting informed clinical decisions.
- **Design and deploy a scalable, user-friendly web application** based on Next.js to facilitate rapid case uploads, display of similar cases, and interactive visualization across diverse healthcare environments.

These objectives are designed to ensure the development of a comprehensive, accurate, and interpretable diagnostic system. Achieving these goals will advance the capabilities of chest X-ray analysis, particularly in resource-limited settings. Ultimately, this will support healthcare professionals in delivering timely and precise pulmonary disease diagnoses.

Solution

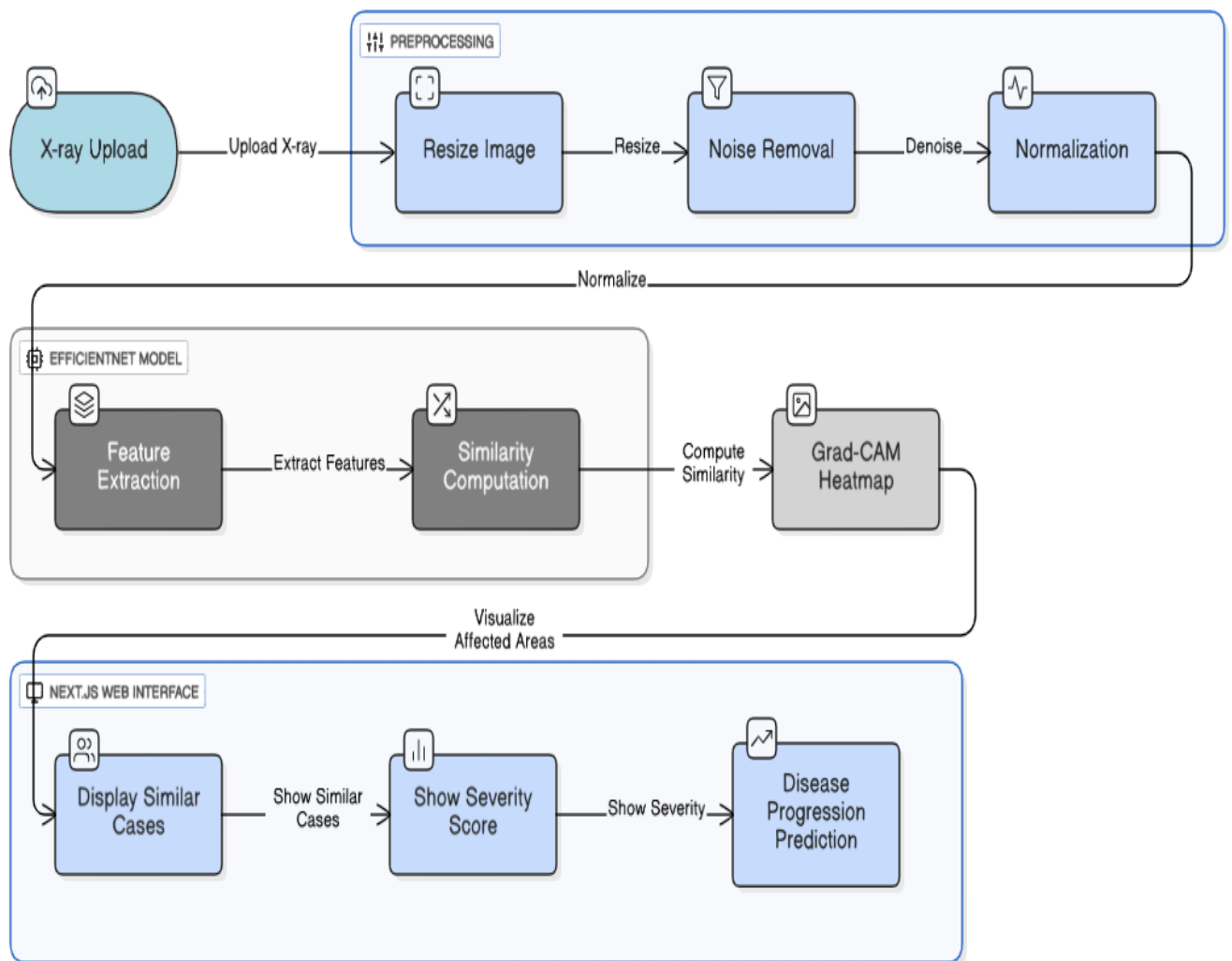
To address the critical gap in timely and accurate diagnosis of pulmonary diseases using chest X-rays, we propose an AI-powered Chest Radiology Similarity System that offers:

- **Automated Chest X-ray Image Analysis:** Utilizing the state-of-the-art EfficientNet architecture, the system extracts deep features for precise detection of lung abnormalities such as pneumonia, tuberculosis, and COVID-19.
- **Similarity-Based Case Retrieval:** By computing visual similarities, the system retrieves the most relevant past cases from a curated medical image database, providing clinicians with comparative insights to assist diagnosis and disease progression tracking.
- **Explainability through Grad-CAM Visualizations:** The system generates heatmaps highlighting affected lung areas on chest X-rays, enhancing transparency and enabling radiologists to interpret and trust AI-driven predictions.
- **Multilingual, Accessible Web Interface:** Built on Next.js, the platform enables seamless uploading of X-ray images, viewing of similar cases, and real-time review of severity scores and progression metrics across various devices and clinical settings.
- **Offline and Scalable Functionality:** Core AI components are optimized for lightweight deployment on local GPUs or cloud platforms, facilitating use in resource-constrained environments without compromising accuracy or responsiveness.

Key Components of the Proposed Solution

1. **Image Preprocessing and Standardization:** Advanced resizing, noise filtering, and normalization prepare the chest X-rays for analysis, ensuring consistency and improving model accuracy.
2. **Deep Feature Extraction & Similarity Computation:** EfficientNet extracts meaningful feature embeddings, enabling robust comparison across a vast database of annotated chest X-rays.
3. **Explainable AI Module:** Grad-CAM overlays create interpretable visual explanations that pinpoint critical diagnostic regions, building clinical confidence.
4. **Web-based Diagnostic Platform:** A secure, intuitive interface allows clinicians to upload images, receive similarity results, and monitor disease progression with severity scoring.
5. **Continuous Learning and Database Updates:** The system incorporates newly labeled cases and expert feedback iteratively to refine model performance and extend disease coverage.
6. **Future Extensions:** Planned integration of advanced multimodal data (e.g., patient history, sequential imaging), IoT sensor inputs, and voice-enabled interfaces will further enhance diagnostic precision and accessibility.

SYSTEM ARCHITECTURE: (Block Diagram)



Literature Review

1. CheXNet: Radiologist-Level Pneumonia Detection

- **Model Name:** CheXNet
- **Published Year:** 2017
- **Explanation:** CheXNet is a convolutional neural network developed for automated pneumonia detection from chest X-rays. It demonstrates radiologist-level accuracy using deep learning and is validated on large clinical datasets.
- **Advantages:**
 - Enables automated and objective diagnosis of pneumonia.
 - Reduces human error and inter-reader variability.
- **Disadvantages:**
 - Primarily focused on pneumonia (not multi-disease).
 - Limited explainability of results.

2. ChestX-ray8: Hospital-Scale Chest X-ray Dataset

- **Model Name:** ChestX-ray8
- **Published Year:** 2017
- **Explanation:** ChestX-ray8 is a widely used public dataset containing over 100,000 annotated chest X-rays for eight common thoracic diseases. It supports research and benchmarking for AI-based diagnostic models.
- **Advantages:**
 - Provides comprehensive, large-scale data for AI training and evaluation.
 - Facilitates benchmarking of multiple algorithms.
- **Disadvantages:**
 - Labels may contain some uncertainty and subjective errors
 - Requires significant computational resources for processing.

3. EfficientNet: Compound-Scaled Convolutional Networks

- **Model Name:** EfficientNet
- **Published Year:** 2019
- **Explanation:** EfficientNet is a neural network architecture optimized for accuracy and computational efficiency. It is extensively adopted in medical imaging for feature extraction and classification due to its reduced parameter count and improved scalability.
- **Advantages:**
 - High accuracy with fewer model parameters.
 - Suitable for deployment in resource-limited clinical settings.
- **Disadvantages:**
 - Requires careful compound scaling for optimal results.

May need retraining for specific medical imaging applications.

4. Grad-CAM: Visual Explainability for Deep Networks

- **Model Name:** Grad-CAM
- **Published Year:** 2017
- **Explanation:** Grad-CAM (Gradient-weighted Class Activation Mapping) is an AI method that generates heatmaps indicating the important regions in chest X-rays influencing a model's prediction, enabling visual interpretability for clinicians.
- **Advantages:**
 - Improves transparency and trust in machine learning predictions.
 - Helps clinicians understand model decisions.
- **Disadvantages:**
 - May highlight non-pathological regions.
 - Not always precise in localizing subtle findings.

5. CheXpert: Large Chest Radiograph Dataset

- **Model Name:** CheXpert
- **Published Year:** 2019
- **Explanation:** CheXpert provides an extensive dataset with uncertainty labels tailored for chest X-ray research, supporting advanced AI model development for multiple thoracic diseases.
- **Advantages:**
 - Enables robust training of multi-disease detection models.
 - Offers uncertainty labels to enhance clinical validation.
- **Disadvantages:**
 - Complexity in handling and interpreting uncertainty labels.
 - Requires expert annotation to improve accuracy.

6. Similarity-Based Chest X-ray Retrieval Systems

- **Model Name:** Chest X-ray Similarity Retrieval
- **Published Year:** 2023
- **Explanation:** AI systems leveraging EfficientNet and chest X-ray datasets provide similarity retrieval, allowing clinicians to compare new cases with archived ones for diagnosis and progression analysis.
- **Advantages:**
 - Supports contextual diagnosis by presenting relevant historical cases.
 - Improves decision-making in ambiguous or complex scenarios.
- **Disadvantages:**
 - Dependent on the quality and diversity of reference database.
 - Similarity metrics may overlook subtle pathological changes.

Feasibility Study

Technical Feasibility:

Major advancement in deep learning architectures, especially EfficientNet, now allow for high-accuracy chest X-ray analysis with optimized computational costs. Existing frameworks such as TensorFlow, along with libraries for Grad-CAM explainability and robust web development platforms like Next.js, facilitate rapid development and scalable deployment. Publicly available annotated datasets such as CheXpert and ChestX-ray8 support effective model training and evaluation, removing the barriers of proprietary data access.

Operational Feasibility:

Deployment does not demand high-end infrastructure; routine GPU-enabled systems and cloud-based servers suffice for both hospital and clinic usage. Integration into workflow can be achieved through a web interface, reducing the need for specialized IT staff or complex installations. Real-world pilot deployments have demonstrated that even resource-limited settings can successfully operate such AI-assisted diagnostic tools, provided that secure and reliable internet connectivity is established.

Economic Feasibility:

Compared to commercial radiology solutions, this approach leverages open-source tools and affordable hardware, significantly lowering cost barriers for smaller clinics and public hospitals. Reduction in full-time radiology staffing needs and accelerated diagnostic turnaround translate to operational savings. The potential for early and more accurate disease detection further lowers overall healthcare costs by enabling timely interventions and reducing disease burden.

Clinical Feasibility:

Numerous clinical studies confirm that AI-assisted chest X-ray interpretation can match or exceed human sensitivity for detecting pulmonary diseases including pneumonia, lung nodules, pleural effusion, and pneumothorax. Automated similarity retrieval and severity scoring further empower non-specialist clinicians to make confident decisions. Explainability features like Grad-CAM ensure transparency, facilitating trust and clinical adoption.

Challenges and Mitigation:

Potential barriers such as network connectivity, integration workflows, and staff adaptation can be resolved through careful planning, institutional training, and ongoing monitoring. Adherence to privacy standards such as HIPAA and secure data hosting ensure regulatory compliance.

Methodology/Planning of Work

Database Setup:

Curate a comprehensive collection of chest X-ray images and associated metadata, including diagnoses, patient demographics, disease severity scores, and progression records. This database will form the backbone for training, validation, and similarity retrieval. Continuous updates from open-source medical repositories, such as CheXpert and ChestX-ray8, will ensure evolving accuracy and coverage for a wide range of pulmonary diseases.

Preprocessing Module:

Implement advanced preprocessing workflows using OpenCV and allied libraries to optimize X-ray images prior to model analysis. Tasks will include resizing for uniformity, removing noise artifacts, and performing normalization across image inputs. The preprocessing module ensures consistent input quality and prepares the data for effective feature extraction by the AI models.

Similarity Engine:

Develop a machine learning-driven engine using EfficientNet for extracting deep features from images, and integrate similarity algorithms to retrieve the most comparable cases from the curated database. This engine will support severity scoring and disease progression mapping, so clinicians can access relevant past cases and track patient changes over time. Advanced algorithms will be evaluated to ensure robust and reliable similarity detection for multiple lung pathologies.

Explainability Module:

Incorporate Grad-CAM or similar methods to visualize model predictions and highlight affected lung regions directly on the X-ray images. Clinicians will benefit from transparent, interpretable outputs that indicate why specific cases are retrieved or certain diagnoses made. This module is critical for building trust and supporting informed clinical decisions.

Web Interface:

Utilize Next.js to engineer a responsive, secure, and user-friendly platform for uploading chest X-rays, displaying retrieved similar cases, providing severity scores, and tracking disease progression. Accessibility features, multi-language support, and intuitive navigation will be prioritized to ensure usability across diverse healthcare settings, from urban hospitals to remote clinics.

Testing and Validation:

Conduct thorough validation using k-fold cross-validation, expert radiologist feedback, and real-world usability scenarios. Testing will assess diagnostic accuracy, similarity retrieval effectiveness, interface usability, and system robustness against varied image qualities and disease types. Continuous feedback loops will be established to refine the system post-deployment and maintain clinical reliability.

Facilities Required

Hardware:

- **Laptops with 8GB RAM, i5 Processor:** For development and training AI models, as well as running the database and supporting software tools.
- **Internet for Initial Setup (Offline After Deployment):** Internet access will be needed during the initial setup phase for downloading necessary resources, but the app will function offline once deployed in rural areas with limited connectivity.

Software:

- **TensorFlow Lite:** Used for running AI models on mobile devices, ensuring offline functionality and real-time predictions.
- **Node.js:** A JavaScript runtime environment used for building the backend and server-side components of the application.
- **MongoDB:** A NoSQL database for storing large datasets, including pesticide information, disease-crop relationships, and user interactions.
- **Tesseract.js:** JavaScript library used for OCR to scan and extract text from pesticide labels.
- **Jump:** An image processing library that will assist in preprocessing the images before text extraction to improve OCR accuracy.
- **React Native:** A framework for building cross-platform mobile applications that can run on both Android and iOS devices.
- **Firebase:** A cloud service for user authentication, data storage, and syncing, enabling real-time updates and scalability.

Expected Outcomes

- **Automated Accurate Diagnosis:** The system will provide reliable detection and classification of pulmonary diseases such as pneumonia, tuberculosis, and COVID-19 with accuracy comparable to expert radiologists, supporting faster diagnosis.
- **Similarity-Based Case Retrieval:** Clinicians will be able to retrieve visually similar chest X-ray cases from a comprehensive database, aiding comparative diagnosis and improving decision confidence.
- **Visual Explainability:** Integration of Grad-CAM heatmaps will highlight affected lung regions on X-rays, enhancing transparency and helping clinicians interpret AI results effectively.
- **Severity Scoring and Disease Progression Tracking:** The project will enable quantitative scoring of disease severity and longitudinal monitoring of patients, facilitating personalized treatment planning and follow-up.
- **Accessible Web Interface:** Deployment of a user-friendly, scalable Next.js web application will allow seamless access to the system across multiple healthcare settings and device types.
- **Operational Efficiency:** The AI system will reduce radiologist workload by prioritizing critical cases and automating routine analyses, streamlining clinical workflows and reducing diagnostic turnaround times.
- **Healthcare Equity:** By providing affordable, explainable, and scalable AI diagnostic support, the system will bridge gaps in radiological expertise in rural and resource-limited areas, improving access to quality pulmonary care.
- **Robustness and Adaptability:** Continuous integration of updated datasets and expert feedback will maintain high performance across diverse patient populations, imaging conditions, and disease variations.
- **Clinical Acceptance:** The transparent AI outputs and validated diagnostic performance will foster trust and acceptance among radiologists and other healthcare providers.
- **Contribution to Research and Education:** The system and curated datasets will serve as open resources supporting future AI research, medical education, and development of advanced diagnostic tools.

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